AI for Medicine – Project Report

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Academic Year: 2025/2026

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Degree Program: Artificial Intelligence

Submission Date: August 2026

1. Project Title

Brain Age Prediction from Multi-Site MRI: Quantifying and Mitigating Site Effects with ComBat-GAM Harmonization

2. Problem Statement

Machine learning models for neuroimaging are increasingly trained on data pooled from multiple hospitals, scanners, and acquisition protocols, since no single site can collect enough subjects on its own. Pooling, however, introduces a well-documented confound known as a batch effect: a systematic, non-biological difference between groups caused by a technical factor of acquisition — here, which scanner or site produced the MRI — rather than the biological variable of interest. The term originates in genomics (batch-processed microarray/RNA-seq data show artificial inter-batch differences unrelated to biology) and applies directly here: different scanners, field strengths, and reconstruction pipelines leave a detectable technical fingerprint on extracted morphometric features, superimposed on the true biological signal.

Left uncorrected, this can bias downstream models, distort apparent group differences, and undermine generalization to a hospital or scanner not seen during development — directly relevant clinically, since a deployed model must generalize beyond its training scanners. Harmonization methods such as ComBat were developed to statistically remove site-related variance while preserving biologically meaningful covariates such as age and sex.

3. Objective of the Study

This study has two related objectives, using regional cortical thickness (CT) and fractal dimension (FD) features extracted from T1-weighted brain MRI across a large, publicly available multi-site dataset:

- Predict chronological brain age from CT/FD features using Leave-One-Site-Out cross-validation (LOSO-CV) with three regression models (ElasticNet, Random Forest, XGBoost).
- Quantify the magnitude of the multi-site batch effect (via a site classifier and PCA visualization) and measure how much ComBat-GAM harmonization reduces it and improves cross-site age-prediction

generalization, evaluated both on development-cohort LOSO-CV and on a genuinely held-out set of sites never used in any development decision.

A specific methodological goal, motivated by course material on reproducibility and data leakage in AI for Medicine, was to design a pipeline that is transparent about every point at which leakage could occur — including subtler forms such as pipeline-selection optimism — rather than only guarding against the most obvious form (train/test contamination within a single split).

4. Dataset Description

The dataset used is the "Brain MR T1-weighted Feature Datasets" released by Marzi, Giannelli, Barucci, Tessa, Mascalchi, and Diciotti (2024) in Scientific Data, archived on Zenodo in two parts, and downloaded programmatically at runtime (not redistributed in the project repository).

Table 1. Dataset composition (subject counts verified programmatically at load time).

Part	Zenodo record	Subjects	Source cohorts
Part I	10.5281/zenodo.7845311	1,190	ABIDE I/II, FCP-ICBM, CoRR-NKI2
Part II	10.5281/zenodo.7845361	550	IXI
Combined	—	1,740	36 distinct acquisition sites

Each row corresponds to one subject (one T1-weighted scan) and contains: a unique Subject identifier, the acquisition SITE, chronological Age, Sex (0 = male, 1 = female), and 22 regional morphometric features — 11 cortical thickness (CT) and 11 fractal dimension (FD) measures, each covering the whole cortex plus four lobes (frontal, temporal, parietal, occipital) in both hemispheres. The features are already-extracted, region-level summary statistics, not raw images. No missing values and no duplicate subject IDs were found (verified with explicit assertions in the notebook).

The dataset combines a mix of clinical (ABIDE: autism spectrum cohorts, predominantly children and young adults) and non-clinical, broader-age cohorts (FCP-ICBM, CoRR-NKI2, IXI). This composition proved consequential: the exploratory analysis in Section 5 shows that the 36 sites differ substantially not only in scanner/protocol but also in age distribution, which has direct implications for how harmonization can be expected to perform (Section 9).

5. Data Preprocessing

Preprocessing consisted of merging the two Zenodo parts (identical column schema, verified programmatically), selecting the 22 CT/FD columns as the feature set (plus Sex as an additional predictor), and standardizing features (zero mean, unit variance) prior to any distance- or variance-sensitive step (PCA, site classifier). No raw images required preprocessing, since the dataset provides already-extracted regional features. An exploratory data analysis (EDA) was performed on the full 1,740-subject cohort — before any train/test or model-selection decision, so it could not introduce leakage — to characterize the batch effect and inform downstream methodological choices.

Key EDA findings:

- Visible and quantified batch effect: a boxplot of whole-cortex CT across the 12 most populous sites, and a 2-D PCA of all 22 standardized features colored by site, both show clear site clustering. One-way ANOVA of each component on SITE (development cohort, raw features): η^2(PC1) = 0.730, η^2(PC2) = 0.671 — higher than the Age–SITE confound below. PC1 separates the IXI/ICBM family from ABIDE; PC2 separates ABIDE-I from ABIDE-II (two data-collection phases years apart, different scanners). PCA clustering alone cannot distinguish a removable technical effect from an unremovable Age confound correlated with site — both look identical. Recomputing η^2 after ComBat-GAM (Section 8) resolves this: η^2(PC2) collapses to 0.077 (removable technical variance, as expected), while η^2(PC1) barely moves to 0.683 (harmonization preserves Age-linked signal by design). PC2 is thus genuine scanner/protocol effect; PC1 is mostly the Age–SITE confound below, not an independently removable technical effect.
- Age–site confound: one-way ANOVA of Age on SITE gives $F = 115.0$, $p \approx 0$, $\eta^2 = 0.703$ — 70.3% of Age variance is explained by site alone. Harmonization needs batch (site) and the preserved covariate (age) to be reasonably independent to fully disentangle them; this dataset does not fully satisfy that assumption.
- Age-range overlap between sites: of all 630 site pairs (36 sites), 142 (22.5%) share zero age-range overlap, giving ComBat no empirical basis there to distinguish a scanner effect from an age effect. IXI-IOP and ABIDEII-BNI_1 — later found to dominate final held-out error (Section 8.4) — are among the most age-isolated sites (zero overlap with 14/35 others each), linking this EDA finding directly to the downstream generalization results.
- Feature multicollinearity: 124/231 feature pairs (53.7%) show $|r| > 0.8$, expected given anatomical redundancy. Whole-cortex CT and FD are strongly correlated (Pearson $r = 0.831$; Spearman $\rho = 0.850$, indicating a mostly, not perfectly, linear relationship). Since Pearson and a linear ablation both assume linearity, the CT/FD ablation was re-run with an additive spline per feature: single-feature R^2 rises once curvature is modeled (CT: 0.566 → 0.661; FD: 0.686 → 0.734), but the gain from combining CT+FD over the best single feature is essentially identical either way (+0.014 both) — the complementary, non-redundant contribution is robust to the linearity assumption, not an artifact of it.
- Non-linear feature–age relationship: linear vs. flexible-spline regression for feature ~ Age, 5-fold cross-validated R^2, shows the spline consistently wins (cortex_CT: 0.570 → 0.642, +0.072; cortex_FD: 0.689 → 0.774, +0.085), motivating ComBat-GAM over standard ComBat (Section 8). Since this pools all 30 sites, a robustness check repeated it within four individual sites (no cross-site mixing). The advantage survives only where the age range extends into childhood/adolescence (ABIDEI-NYU, 6.5–31.8y: +0.075 R^2 for CT) and vanishes in adult-only sites (IXI-Guys, IXI-HH, both 20+: −0.018, −0.058) — consistent with cortical thickness changing rapidly in development and near-linearly in adulthood, confirming genuine biological curvature, not a pooling artifact.

6. Avoiding Data Leakage

Three distinct forms of potential leakage were identified and addressed explicitly:

(a) Standard train/test leakage in the predictive model.

Age prediction is evaluated with Leave-One-Site-Out cross-validation (LOSO-CV): for each of the 30 development sites in turn, all subjects from that site are held out entirely, the model is trained on the remaining sites, and predictions are made only on the held-out site. Hyperparameter tuning (RandomizedSearchCV, 20 iterations) is nested strictly inside each outer training fold using an inner 3-fold KFold split, so the held-out site never influences model or hyperparameter selection.

(b) An architectural limitation of ComBat harmonization for LOSO-CV.

The original design attempted to fit ComBat separately inside each LOSO fold, learning batch parameters only on the training sites and applying them out-of-sample to the held-out site. This fails in practice, confirmed directly from the `neuroHarmonize` source (`harmonizationApply.py`): the function checks whether each subject's SITE was present in the original fit (`isTrainSite = covars['SITE'].isin(model['SITE_labels'])`), and for any subject whose site was not — exactly the held-out site in every LOSO fold — it returns `NaN` for that subject's harmonized features rather than a usable value; there is no parameter to handle this differently. `NaN` features are unusable by any downstream regressor, so a strictly per-fold ComBat fit cannot produce a working held-out prediction. This reflects a genuine, general limitation of ComBat-style harmonization — correcting data from a site with zero prior reference subjects is an open problem — not merely a quirk of this implementation. The workaround adopted, standard in the harmonization literature, is to fit ComBat-GAM once, globally, on the full development cohort, then run LOSO-CV for the predictive model on top of the harmonized features. The predictive model's LOSO-CV integrity is fully preserved (a held-out site's true age is never used to fit or tune the regression models); what differs from a strictly out-of-sample design is that the unsupervised harmonization step uses every development subject's age, including subjects in what will later be each LOSO test fold, to estimate the shared age–feature relationship it should preserve. This is reported transparently as a limitation rather than silently worked around.

(c) Pipeline-selection leakage.

A subtler risk: even with correct LOSO-CV, choosing which harmonization variant or model to prefer by watching the same metric later reported as the final result introduces mild optimism. To guard against this, six entire sites (261 subjects) were held out at random (seed=42) immediately after the EDA, before any harmonization-method, model, or hyperparameter decision, and touched only once, at the end, for a single final evaluation (Section 8). The split is site-level, not subject-level, since random subjects would still let the model implicitly "see" a held-out site's scanner signature through other subjects of the same site in training. The gap between the LOSO-CV estimate and this final estimate directly quantifies the pipeline-selection optimism.

(d) Inner hyperparameter-tuning CV leakage.

The inner CV (Section 7) is a plain KFold (k=3), not grouped by site — subjects from the same training site can land in both the inner-train and inner-validation split within a given hyperparameter trial. This is milder than (a): it can bias which hyperparameters are selected, but never leaks the true outer-test site, which is excluded from all 29 training sites regardless of inner-CV design, so reported generalization numbers are unaffected. This was tested, not assumed: grouping the inner CV by site (`GroupKFold`) was found to trade this mild leakage for a worse problem, a direct consequence of the Age–SITE confound ($\eta^2 = 0.703$, Section 5). Only 2 of the 30 development sites (`IXI-Guys`, `IXI-HH`) contain any subject aged 60+; with site-grouping, one inner fold received zero such subjects (60/48/49 per fold ungrouped vs.

103/54/0 grouped), leaving hyperparameter selection with no validation signal for older subjects. This is structural, not a choice of k: for the 2/30 outer folds where the held-out site is itself `IXI-Guys` or `IXI-HH`, only one older-adult site remains in training, so any site-grouped inner split necessarily leaves a fold blind to that age group. The ungrouped design was kept as the better-understood trade-off.

7. Machine Learning Pipeline

Three regression models were compared for age prediction, each with a RandomizedSearchCV hyperparameter space (20 iterations, scored by negative MAE, 3-fold inner KFold):

Table 2. Model space and tuned hyperparameters.

Model	Key hyperparameters searched
ElasticNet	alpha (log-scale, 30 values), l1_ratio (19 values, 0.05–0.95)
Random Forest	n_estimators, max_depth, min_samples_leaf, max_features
XGBoost	n_estimators, max_depth, learning_rate, subsample, colsample_bytree

Validation strategy: LOSO-CV as the outer loop (Section 6), nested RandomizedSearchCV as the inner loop. Evaluation metrics: mean absolute error (MAE) and root mean squared error (RMSE) in years, and R^2. Because per-site R^2 can be misleadingly strongly negative for sites with a narrow internal age range (r2_score compares errors against that site's own small variance, not the global one — a statistical artifact of averaging R^2 across heterogeneous groups, not a modeling error), a pooled R^2 — computed on all out-of-fold predictions concatenated together — is reported alongside the per-site-averaged figures as the more stable aggregate. A separate diagnostic Random Forest classifier (balanced class weights, stratified 5-fold CV) is used to quantify site separability directly, both before and after harmonization, reported as balanced accuracy and macro-F1 against the chance level 1/(number of sites).

8. Results

8.1 Site separability (batch effect quantification)

On raw (unharmonized) features, the site classifier achieves balanced accuracy = 0.385 and macro-F1 = 0.398 on the 30-site development cohort, against a chance level of ≈0.033 (1/30) — an 11.6× improvement over random guessing, direct confirmation of a strong, real batch effect.

After ComBat-GAM harmonization, site separability drops sharply: balanced accuracy falls to 0.099, macro-F1 to 0.088 (−74%), confirming ComBat-GAM removes most site-related signal. The residual (0.099) stays ≈3× above chance, consistent with the age–site confound ($\eta^2 = 0.703$, Section 5): full removal of site information is not expected while age and site remain this entangled.

8.2 Age prediction: development-cohort LOSO-CV, raw vs. harmonized

Table 3. LOSO-CV performance, mean ± SD across the 30 development sites. R^2 is per-site-averaged and known to be unstable for narrow-age-range sites (see Section 7); pooled R^2 for the best model is reported below.

Model	Condition	MAE (yr)	RMSE (yr)	R^2 (per-site mean)
ElasticNet	Raw	6.97 ± 3.42	8.46 ± 3.87	−11.74 ± 24.05
ElasticNet	Harmonized	6.50 ± 1.74	8.20 ± 2.12	−10.04 ± 12.81
Random Forest	Raw	4.84 ± 3.61	6.40 ± 4.50	−5.86 ± 21.09
Random Forest	Harmonized	3.83 ± 2.19	5.55 ± 3.02	−1.58 ± 2.68
XGBoost	Raw	4.68 ± 3.21	6.20 ± 4.10	−3.87 ± 11.26
XGBoost	Harmonized	3.82 ± 2.25	5.44 ± 3.11	−1.51 ± 2.72

XGBoost achieves the lowest mean LOSO MAE under harmonization (3.82 years) and was selected as the best model for downstream use. Harmonization improves mean MAE for every model. On pooled out-of-fold predictions, the harmonized XGBoost model reaches MAE = 5.05, RMSE = 7.85, R^2 = 0.839, versus a pooled raw baseline of MAE = 5.95, R^2 = 0.771 (ElasticNet: 8.70/0.644 → 7.91/0.705; Random Forest: 6.07/0.768 → 5.03/0.837) — confirming the same improvement pattern seen per-site above.

The improvement is not evenly distributed, however. Counting per-site wins/losses: ElasticNet improves on only 11/30 sites (median change +0.35y, worse for the typical site); Random Forest 16/30 (−0.02y, a wash); XGBoost 16/30 (−0.27y, the most consistent). The pooled/mean improvement is driven disproportionately by a subset of sites with large gains — plausibly the wide-age-range, strongly-confounded sites identified in Sections 5 and 8.4.

A paired Wilcoxon signed-rank test (per-site MAE, harmonized vs. raw) does not reach conventional significance for any model: ElasticNet p = 0.903, Random Forest p = 0.309, XGBoost p = 0.096 — all improving in direction, none conclusive alone, given only 30 paired observations and substantial site heterogeneity. The permutation-based robustness check (Section 8.5), which isolates the harmonization mechanism directly, remains the stronger evidence the effect is genuine.

8.3 Vanilla ComBat vs. ComBat-GAM

An initial run with standard (linear-covariate) ComBat increased MAE/RMSE and decreased pooled R^2 for every model — the opposite of the intended effect. This matches the non-linear feature–age relationship from Section 5: forcing a linear term onto a genuinely curved relationship lets ComBat distort real biological variance along with the site effect it should remove. Switching to ComBat-GAM (`smooth_terms=["Age"]`, a B-spline instead of a rigid linear coefficient) resolved this, producing the expected improvement reported above.

8.4 Final held-out evaluation (sites never used in any development decision)

Table 4. XGBoost, best model selected using development-cohort results only. Final held-out set: 6 sites, 261 subjects, never used in any harmonization-method, model, or hyperparameter decision. ComBat-GAM was re-fit once, jointly over the development cohort and the final-test sites, after every modeling decision was already locked using development-only results (mirrors a realistic deployment scenario: harmonizing a newly arrived site against the reference cohort).

Evaluation	MAE (yr)	RMSE (yr)	R^2
Development-cohort LOSO-CV, harmonized (pooled)	5.05	7.85	0.839
Final held-out sites, raw	10.09	14.08	0.445
Final held-out sites, harmonized	7.57	10.57	0.687

Harmonization improves final held-out performance substantially (MAE 10.09 → 7.57 years, a 25% reduction; R^2 0.445 → 0.687), confirming the benefit generalizes to genuinely unseen sites and is not an artifact of pipeline-selection on the development cohort. The gap between the development-cohort pooled estimate (R^2 = 0.839) and the final held-out estimate (R^2 = 0.687) is a direct, quantified measure of the pipeline-selection optimism discussed in Section 6: even with correct per-fold LOSO-CV, a pipeline iteratively refined by watching the same metric carries some unavoidable optimism, and this study measured it explicitly rather than ignoring it.

A per-site breakdown shows the pooled error is not evenly distributed: two sites, ICBM (n=86, age 19–85) and IXI-IOP (n=63, age 20–86), account for 76% of the sample-weighted MAE despite only 57% of final-test subjects, while the four ABIDE-derived sites (younger, narrower ranges) predict much more accurately (MAE 2–4y). Harmonization helps most at ICBM (MAE 15.44 → 9.71y); IXI-IOP is essentially unchanged (10.97 → 11.13y).

Individual predictions reveal a systematic age-dependent bias: the oldest subjects are consistently under-predicted (e.g. true 85 → predicted 67.2y; true 68 → predicted 47.8y), while younger-subject errors are small and non-directional — the well-documented regression-to-the-mean effect in brain-age prediction (Cole & Franke, 2017), driven by the young-skewed (ABIDE-heavy) training distribution. This directly explains part of the elevated error at ICBM and IXI-IOP.

8.5 Robustness check: is the harmonization benefit a covariate-preservation artifact?

ComBat-GAM is given each subject's true Age to fit a smooth population-level curve it must preserve while removing the site effect. This raises a legitimate concern: the improvement above could reflect mechanical shrinkage toward a smooth age-conditional curve — leakage via preprocessing — rather than genuine site-noise removal, regardless of whether the covariate carries real information.

To test this, ComBat-GAM was re-fit with Age replaced by a random permutation across the development cohort (SITE left untouched), and the same LOSO-CV pipeline predicted each subject's true Age from these permuted-covariate-harmonized features.

Table 5. Permuted-covariate control, development cohort, pooled out-of-fold predictions.

Model	Raw (pooled)	True-covariate harmonized (pooled)	Permuted-covariate harmonized (pooled)
ElasticNet	MAE 8.70 / R^2 0.644	MAE 7.91 / R^2 0.705	MAE 17.02 / R^2 −0.136

Random Forest	MAE 6.07 / R^2 0.768	MAE 5.03 / R^2 0.837	MAE 14.67 / R^2 0.087
XGBoost	MAE 5.95 / R^2 0.771	MAE 5.03 / R^2 0.841	MAE 14.15 / R^2 0.113

If the benefit were a covariate-preservation artifact, the permuted-covariate condition should resemble the true-covariate one (both better than raw). Instead it is far worse than raw for every model (MAE 2.4–2.9× higher; R^2 near zero or negative) — a meaningless covariate actively destroys signal, spending the correction budget removing deviation from a fictitious curve rather than a genuine site effect. This rules out the artifact explanation: the improvements in Sections 8.2–8.4 reflect genuine site-effect removal, conditional on the covariate being real.

8.6 Age-bias correction (addressing the regression-to-the-mean effect)

Section 8.4 showed systematic under-prediction of the oldest subjects. Following Smith et al. (2019), a linear correction (predicted_age ~ true_age) was fit on development-cohort, harmonized, out-of-fold XGBoost predictions only, then inverted and applied to final held-out predictions without using their true age — no leakage from held-out labels.

The fit: predicted_age = 4.16 + 0.827 × true_age. $\beta = 0.827 < 1$ confirms regression-to-the-mean: predictions compress toward the training mean age. The inverse correction, corrected_age = (predicted_age − 4.16) / 0.827, was applied to final held-out predictions.

Table 6. XGBoost, final held-out set (6 sites, 261 subjects). Bias-correction parameters ($\alpha = 4.16$, $\beta = 0.827$) fit exclusively on development-cohort out-of-fold predictions; never refit or tuned on the final held-out set.

Metric	**Harmonized, uncorrected**	**Harmonized, bias-corrected**
MAE (yr)	7.54	8.48
RMSE (yr)	10.50	11.56
R^2	0.691	0.626
Correlation(true age, signed error)	r = −0.422 (p = 1.05e-12)	r = −0.119 (p = 0.054)

Point-accuracy metrics get modestly worse after correction — the expected, documented trade-off (Smith et al., 2019; de Lange & Cole, 2020), not a failure: dividing by $\beta = 0.827$ removes systematic bias but also amplifies residual random error by the same factor (≈1.21×). What matters for this correction's purpose is the age–error correlation, since a brain-age gap still correlated with age is a confound for any downstream biomarker use — different-age groups would show a spurious gap difference regardless of real biological effect. Here the correlation drops >3.5× and loses conventional significance ($r = -0.422$, $p<0.001$ → $r = -0.119$, $p=0.054$), confirming the correction works as intended. Both metrics are reported so the appropriate version can be chosen by downstream goal (raw accuracy vs. age-independent gap).

9. Discussion

Taken together, the results support four main conclusions. First, the multi-site batch effect is real and substantial, not a modeling artifact: visible qualitatively (PCA, boxplots), quantifiable directly (site classifier far above chance), and consequential for downstream prediction. Second, the harmonization method matters as much as whether to harmonize at all: standard linear-covariate ComBat measurably hurt performance, and only modeling the non-linear age–feature relationship (ComBat-GAM) recovered the expected benefit — anticipated by EDA evidence (Section 5) before it was observed downstream, strengthening confidence it reflects a genuine data property rather than post-hoc rationalization. Third, honest evaluation matters: the gap between the optimistic development-cohort estimate (R^2 = 0.839) and the genuinely held-out estimate (R^2 = 0.687) would have gone unmeasured without the dev/final-test site-level split adopted specifically to guard against pipeline-selection optimism. Fourth, the harmonization benefit is not a covariate-preservation artifact: the permuted-covariate control (Section 8.5) shows a meaningless Age covariate produces performance far worse than not harmonizing at all, confirming the benefit depends on a genuine Age–feature relationship.

Limitations

- Age–site confound (η^2 = 0.703): age and site are strongly entangled in this cohort, which structurally limits how completely any harmonization method can disentangle scanner effects from genuine age-related variance. This is a property of the dataset, not an implementation flaw, but it means the harmonization results should not be read as a fully clean separation of the two sources of variance.
- Regression-to-the-mean age bias, most visible on the final held-out set, is a known limitation of brain-age models trained on age-imbalanced cohorts: only 10.8% of subjects (188/1,740) are aged 60+, from just 5 of the 36 sites, and two sites alone (IXI-Guys, IXI-HH) account for 83.5% of that group. A dedicated bias-correction step (Section 8.6, Smith et al., 2019) removes most of the age-dependence in the error (correlation with true age: r = −0.422 → −0.119), at the cost of somewhat worse point-accuracy metrics — a documented trade-off, not a failure; the root cause remains the age-composition imbalance itself, which no post-hoc correction fully substitutes for genuine site diversity. Naive oversampling (e.g. SMOGN-style resampling) was rejected, since interpolating within the existing sparse, site-homogeneous older-adult pool would not add genuine site diversity.
- The final held-out MAE is dominated by two sites with the widest age range (ICBM, IXI-IOP); with only 6 held-out sites, this estimate has non-trivial site-to-site variance and should not be over-interpreted as a single precise generalization number.
- ComBat is fit globally on the development cohort rather than strictly out-of-sample per LOSO fold, for the architectural reasons explained in Section 6; this is a standard, documented simplification in the harmonization literature, but is not a fully leakage-free design for the harmonization step specifically (the predictive model's own LOSO-CV integrity is unaffected).
- The inner hyperparameter-tuning CV is not grouped by site (Section 6d) — a deliberate, tested trade-off, not an oversight: grouping was found to leave some folds with zero older-adult validation signal, a worse problem than the mild tuning-selection leakage it would remove.
- The brain-age gap (predicted − chronological age), a biomarker associated with neurodegenerative risk in the literature (Cole & Franke, 2017), could not be clinically validated on this dataset, since none of the source cohorts (ABIDE, FCP-ICBM, CoRR-NKI2, IXI) include neurodegenerative-disease diagnoses or an elderly clinical cohort.
- Unmodeled diagnostic status in the ABIDE-derived sites: by design, both ABIDE I (539 ASD / 573 typically-developing controls, ages 7–64) and ABIDE II (521 ASD / 593 controls, ages 5–64) are case-control initiatives, roughly balanced between autism-spectrum and typically-developing subjects. The harmonization dataset used here (Marzi et al., 2024) carries no diagnostic-group column, so ASD

status could neither be verified, controlled for, nor reported for the ABIDE-derived sites, which make up the majority of the cohort. Findings on structural cortical differences in ASD are heterogeneous in the literature, so the practical impact is unclear, but it is an additional source of biological variance, distinct from age and sex, that this study could not account for — and if diagnostic composition varies by site, it could itself contribute to part of the site-related variance attributed to scanner/protocol effects.

10. Conclusions and Future Work

This study demonstrates a real, substantial multi-site batch effect in a 36-site, 1,740-subject brain MRI feature dataset, shows that ComBat-GAM harmonization measurably improves cross-site brain-age prediction (both in development-cohort LOSO-CV and on a genuinely held-out set of sites), and explicitly quantifies the pipeline-selection optimism that a naive evaluation would have missed ($R^2 = 0.839$ optimistic vs. 0.687 honest held-out estimate). The choice of ComBat-GAM over standard ComBat was empirically necessary, not merely preferable, given the demonstrably non-linear relationship between cortical features and age in this cohort.

Future work:

- Investigate whether the small residual age–error correlation after bias correction ($r = -0.119$, $p = 0.054$, Section 8.6) reflects a genuine, small dev/final-test distribution shift, or is sampling noise at $n = 261$; a repeated held-out-site resampling experiment (see next item) could disambiguate the two.
- Explore a combined stratified-and-grouped inner CV (e.g. jointly respecting site grouping and age-bracket balance) rather than plain `GroupKFold` by site alone — tested directly (Section 9), plain site-grouping was found to trade the mild inner-CV leakage for zero older-adult validation signal in some folds; a design that also balances age brackets might avoid both, though with only 29 training sites this may not be achievable without a larger or more site-diverse older-adult cohort than currently available.
- Extend the final held-out evaluation with more than 6 sites, or repeated random site-level splits, to characterize the variance of the generalization estimate itself.
- Investigate site-level harmonization diagnostics (e.g., per-site residual distributions) to better separate genuine scanner effects from the age–site confound identified in Section 5.

11. Ethics and Data Privacy

All data used are secondary, publicly available, de-identified datasets — only aggregated regional morphometric features are used, never raw images — originally collected under informed consent and institutional ethics approval by the source studies (ABIDE, FCP-ICBM, CoRR-NKI2, IXI). The dataset (Marzi et al., 2024) is archived on Zenodo (records 7845311 and 7845361) and distributed by its original authors under a CC BY(-NC)-SA 3.0 license, restricted to academic/research use; no data are redistributed in this project's repository, and the notebook downloads them directly from Zenodo at runtime. All code developed for this project is released under the MIT license. This project is for academic purposes only and makes no clinical claims; in particular, the brain-age-gap analysis (Section 8) is explicitly reported as descriptive, not as a validated clinical biomarker on this dataset.

12. Code and Reproducibility

The complete, self-contained analysis is implemented in a single Google Colab notebook, AI_for_Medicine_BrainAge.ipynb, which installs its own dependencies and downloads the data directly from Zenodo in its first cells — no manual setup is required to reproduce the results.

- Reproducibility measures: a single global random seed (RANDOM_STATE = 42) fixes NumPy, all model constructors, and all cross-validation splitters. All three regression models (ElasticNet, Random Forest, XGBoost) are deterministic given a fixed seed — no GPU-induced non-determinism applies, unlike deep learning models.
- Dependency pinning: key package versions are pinned in requirements.txt (neuroHarmonize, neuroCombat, xgboost, scikit-learn, etc.).
- Cross-environment validation: in addition to the primary Google Colab environment (pinned dependencies), the full pipeline was independently re-run in a local Python 3.8 environment with unpinned, older-compatible package versions. The final held-out results matched closely (e.g. harmonized XGBoost: MAE 7.57 years on Colab vs. 7.72 years locally; R^2 0.687 vs. 0.675), confirming the pipeline's conclusions are not an artifact of a specific software environment.
- FAIR archiving: the GitHub repository is archived on Zenodo with a permanent DOI (10.5281/zenodo.21904829, release v1.0.0), following the FAIR (Findable, Accessible, Interoperable, Reusable) principles for research software discussed in the course.
- Caching: a RUN_FULL_TRAINING flag allows the full nested LOSO-CV to be reproduced from scratch, or precomputed results to be reloaded from the results/ directory for a fast walkthrough.

13. References

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